

LABORATORY PROJECT REPORT

COLOR DETECTION AND ANALYSIS

EXPERIMENT 8

Section 1

Group 7

Semester 2 2025/2026

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Abstract

This experiment investigates the implementation of an AI-based color recognition system through the integration of the Gravity HuskyLens AI Vision Camera with an Arduino Uno microcontroller platform. The primary objective of the project was to develop a real-time embedded vision system capable of detecting and classifying specific colors while providing immediate visual feedback through an RGB LED (KY-016). The HuskyLens camera was configured and trained using its onboard machine learning-based color recognition algorithm to identify red, green, and blue objects, with each detected color assigned to a unique identification (ID) parameter.

During operation, the HuskyLens transmitted visual recognition data to the Arduino Uno through serial communication, enabling the microcontroller to process the received information and generate the corresponding RGB LED output. Pulse Width Modulation (PWM) control was utilized to regulate the RGB LED channels and accurately reproduce the detected color in real time. The system demonstrates the integration of machine vision sensing, embedded control, serial data interfacing, and actuator-based feedback within a compact mechatronic platform without requiring external image processing software or high-level computing systems.

Through this experiment, students gained practical exposure to AI-assisted vision sensing, real-time signal processing, microcontroller-based hardware interfacing, and intelligent automation concepts. In addition, the project strengthened understanding of mechatronic system integration involving sensor acquisition, communication protocols, decision-making processes, and responsive actuator control based on visual environmental input.

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1.0 Introduction

Image and video interfacing technologies play a significant role in modern mechatronic and embedded systems, particularly in applications involving automation, robotics, intelligent sensing, and machine vision. Conventional color detection systems commonly rely on dedicated color sensors, which are often sensitive to environmental lighting variations and require continuous calibration to maintain detection accuracy. With the advancement of artificial intelligence (AI)-based vision technology, smart vision modules such as the Gravity HuskyLens AI Vision Camera provide a more adaptive, reliable, and efficient solution for real-time visual recognition and object classification tasks.

In this experiment, the Gravity HuskyLens AI Vision Camera was utilized as the primary image acquisition and color recognition device. The HuskyLens employed its onboard AI-based machine vision algorithm to detect and classify predefined colors including red, green, and blue through its built-in color recognition mode. The detected visual information was then transmitted to an Arduino Uno microcontroller through Universal Asynchronous Receiver-Transmitter (UART) serial communication for real-time data processing and decision-making operations.

To provide immediate system feedback, an RGB LED (KY-016) module was integrated as the output actuator. Based on the color identification result received from the HuskyLens, the Arduino Uno generated the corresponding RGB output signal using Pulse Width Modulation (PWM) techniques to illuminate the LED with the detected color. This experiment demonstrates the integration of AI-assisted machine vision sensing, embedded system programming, serial communication interfacing, and real-time actuator control within a compact mechatronic platform without requiring external image processing software or high-performance computing hardware.

Furthermore, the experiment provides practical exposure to intelligent sensing technologies and strengthens understanding of mechatronic system integration involving sensors, communication protocols, signal processing, and responsive output control. The concepts and techniques explored throughout this experiment have potential applications in industrial automation systems, automated product sorting, intelligent traffic management, robotic inspection systems, and interactive human-machine interface (HMI) technologies.

2.0 Materials and Equipment

2.1 Electronic Components

The following electronic components were utilized in the development of the AI-based color recognition system:

- Arduino Uno microcontroller board
- Gravity HuskyLens AI Vision Camera
- KY-016 RGB LED module
- Breadboard
- Jumper wires
- Current-limiting resistors for RGB LED protection

2.2 Equipment and Software Tools

The equipment and software tools required for this experiment are listed below:

- Arduino IDE software for program development and code uploading
- USB cable for Arduino Uno communication and power supply
- HuskyLens built-in display and navigation buttons for AI training and configuration
- Personal computer or laptop for system programming and monitoring

3.0 Experimental Setup

The experimental setup was designed to integrate the Gravity HuskyLens AI Vision Camera with an Arduino Uno microcontroller and an RGB LED module to develop a real-time AI-based color recognition system. The HuskyLens functioned as the primary machine vision sensor for color detection, while the Arduino Uno acted as the embedded processing and control unit. The KY-016 RGB LED module was utilized as a visual output actuator to indicate the detected color in real time.

1. The Gravity HuskyLens AI Vision Camera was interfaced with the Arduino Uno through UART serial communication using the SoftwareSerial library for data transmission between the vision sensor and the microcontroller.
2. The VCC and GND terminals of the HuskyLens were connected to the 5V and GND pins of the Arduino Uno to provide stable operating power for the vision module.
3. The TX and RX communication pins of the HuskyLens were connected to Arduino Uno digital pins D5 and D6 respectively to establish bidirectional serial communication.

4. The KY-016 RGB LED module was connected to PWM-capable digital output pins of the Arduino Uno to enable Pulse Width Modulation (PWM)-based color generation and brightness control:
 - Red channel → D9
 - Green channel → D10
 - Blue channel → D11
5. The Arduino Uno board was powered and programmed through a USB connection to the computer using the Arduino IDE environment.
6. The HuskyLens was configured in Color Recognition Mode, where predefined colors including red, green, and blue were trained using the built-in AI learning function and onboard control buttons. Each trained color was assigned a unique identification (ID) for classification during real-time operation.
7. During operation, the HuskyLens continuously analyzed visual input and transmitted the detected color ID to the Arduino Uno. The Arduino then processed the received serial data and activated the corresponding RGB LED color output in real time.

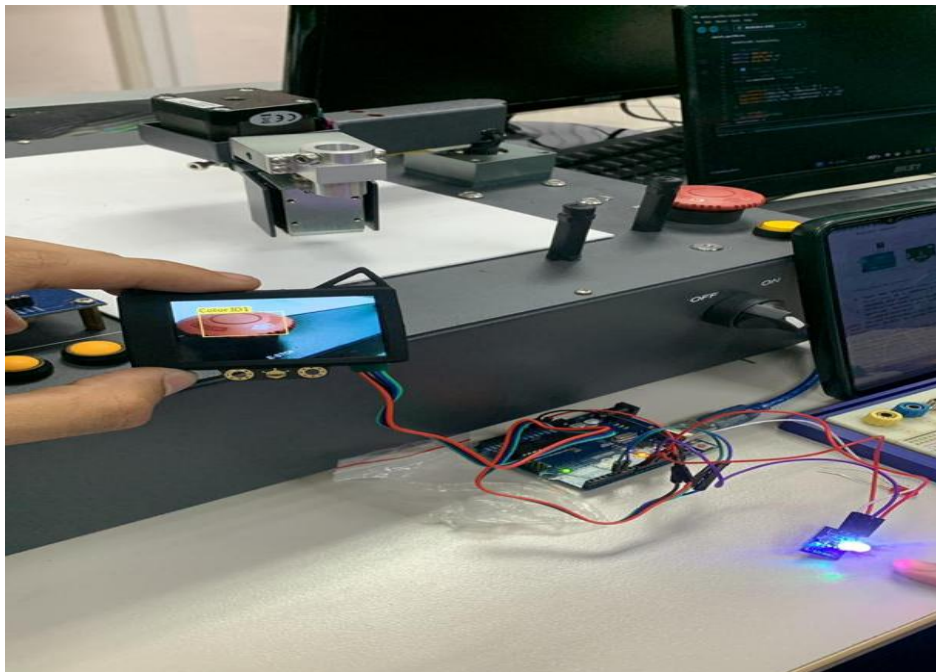
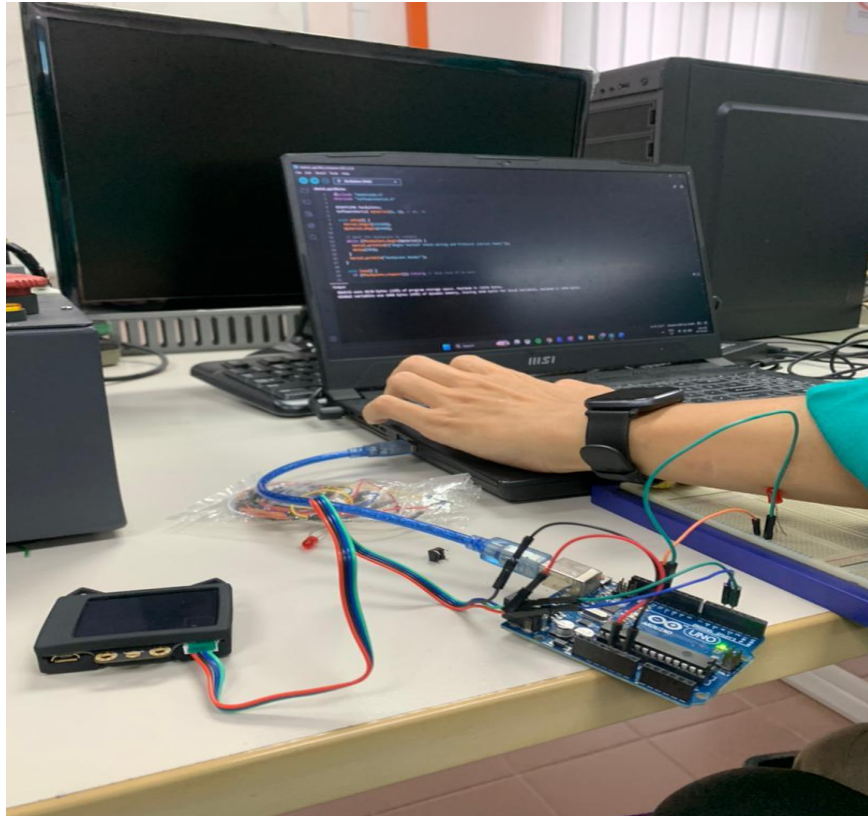


Figure 1 shows the husky lens working



Figures 2 shows the physical connections

4.0 Methodology

The experiment was conducted systematically to ensure accurate AI-based color recognition, reliable serial communication, and stable real-time hardware response between the HuskyLens AI Vision Camera, Arduino Uno, and RGB LED module. The methodology involved hardware configuration, AI training, embedded programming, serial data acquisition, and output verification under different operating conditions.

4.1 Detailed Procedure

1. The Gravity HuskyLens AI Vision Camera was powered ON and configured to operate in **Color Recognition Mode** using the built-in display interface and onboard navigation buttons.
2. Three predefined colored objects consisting of red, green, and blue were individually trained using the HuskyLens machine learning function. Each detected color was assigned a unique identification (ID) for classification during real-time operation.
3. The Arduino Uno microcontroller was connected to the computer through a USB interface, and the control program was developed, compiled, and uploaded using the Arduino IDE software environment.

4. The program initialized UART serial communication between the Arduino Uno and the HuskyLens AI Camera through the **SoftwareSerial** library to enable continuous data exchange between the vision sensor and microcontroller.
5. During operation, the Arduino Uno continuously requested and monitored detection data transmitted from the HuskyLens AI Vision Camera in real time.
6. Once a trained color object was detected, the HuskyLens transmitted the corresponding color ID together with object coordinate information to the Arduino Uno for further processing and output control.
7. Based on the received identification (ID), the Arduino activated the corresponding RGB LED output using Pulse Width Modulation (PWM) control:
 - Red object detected → Red LED channel activated
 - Green object detected → Green LED channel activated
 - Blue object detected → Blue LED channel activated
8. If no trained object or an unrecognized color was detected within the camera frame, all RGB LED outputs were automatically deactivated to indicate the absence of a valid detection signal.
9. The detected color ID, object position coordinates, and system response data were displayed on the Arduino IDE Serial Monitor to facilitate real-time monitoring, debugging, and communication verification.
10. The overall system performance was evaluated under different object distances, viewing angles, and ambient lighting conditions to analyze the consistency, responsiveness, and reliability of the AI-based color recognition system.

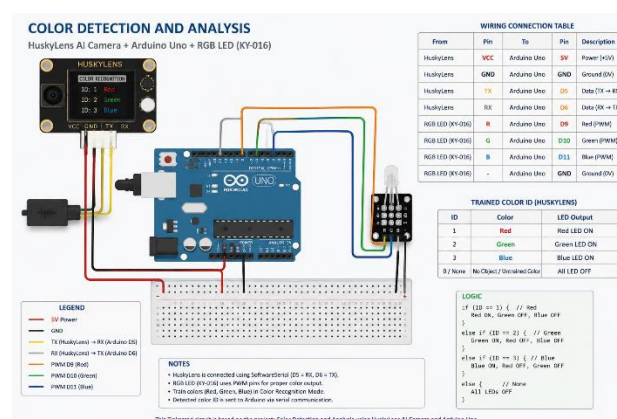


Figure 3 shows the thinkerCad Connection

4.1 Programming

Task 1: Single Object Detection with LED Output

```
#include "HUSKYLENS.h"
```

```
#include "SoftwareSerial.h"
```

```
HUSKYLENS huskylens;
```

```
SoftwareSerial mySerial(4, 5); // RX, TX
```

```
const int ledPin = 13;
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    mySerial.begin(9600);
```

```
    pinMode(ledPin, OUTPUT);
```

```
    while (!huskylens.begin(mySerial)) {
```

```
        Serial.println(F("Begin failed! Check wiring and Protocol (Serial 9600)"));
```

```
        delay(500);
```

```
    }
```

```
    Serial.println("HuskyLens Ready!");
```

```
}
```

```
void loop() {
```

```
    if (!huskylens.request()) {
```

```
        digitalWrite(ledPin, LOW);
```

```
        return;
```

```
    }
```

```
if (huskylens.available()) {  
    HUSKYLENSResult result = huskylens.read();  
  
    if (result.command == COMMAND_RETURN_BLOCK && result.ID == 1) {  
        Serial.println("Object 1 detected!");  
        digitalWrite(ledPin, HIGH);  
    } else {  
        digitalWrite(ledPin, LOW);  
    }  
    } else {  
        digitalWrite(ledPin, LOW);  
    }  
}
```

Task 2: Multi-Color Detection and RGB LED Control

```
#include "HUSKYLENS.h"  
#include "SoftwareSerial.h"
```

```
HUSKYLENS huskylens;  
SoftwareSerial mySerial(4, 5); // RX, TX
```

```
const int redPin = 9;  
const int greenPin = 10;  
const int bluePin = 11;
```

```
void setup() {  
    Serial.begin(115200);  
    mySerial.begin(9600);  
}
```

```
pinMode(redPin, OUTPUT);
```

```
pinMode(greenPin, OUTPUT);
```

```
pinMode(bluePin, OUTPUT);
```

```
clearLED();
```

```
while (!huskylens.begin(mySerial)) {
```

```
    Serial.println(F("Begin failed!"));
```

```
    delay(500);
```

```
}
```

```
Serial.println("HuskyLens Ready for 5 Colors!");
```

```
}
```

```
void loop() {
```

```
    if (!huskylens.request()) {
```

```
        clearLED();
```

```
        return;
```

```
}
```

```
if (huskylens.available()) {
```

```
    HUSKYLENSResult result = huskylens.read();
```

```
    if (result.command == COMMAND_RETURN_BLOCK) {
```

```
        switch (result.ID) {
```

```
            case 1: // RED
```

```
    setColor(255, 0, 0);  
    Serial.println("Detected: Red");  
    break;  
  
case 2: // BLUE  
    setColor(0, 0, 255);  
    Serial.println("Detected: Blue");  
    break;  
  
case 3: // GREEN  
    setColor(0, 255, 0);  
    Serial.println("Detected: Green");  
    break;  
  
case 4: // YELLOW  
    setColor(255, 255, 0);  
    Serial.println("Detected: Yellow");  
    break;  
  
case 5: // PURPLE  
    setColor(255, 0, 255);  
    Serial.println("Detected: Purple");  
    break;  
  
default:  
    clearLED();  
    break;  
}  
}
```

```

    } else {
        clearLED();
    }
}

void setColor(int redValue, int greenValue, int blueValue) {
    analogWrite(redPin, redValue);
    analogWrite(greenPin, greenValue);
    analogWrite(bluePin, blueValue);
}

void clearLED() {
    setColor(0, 0, 0);
}

```

5.0 Data Collection

The data collection process was conducted by testing the HuskyLens AI Vision Camera with five trained colors: red, blue, green, yellow, and purple. Each color was assigned a unique identification ID during the training process. The detected ID was then transmitted to the Arduino Uno through serial communication, allowing the Arduino to activate the corresponding RGB LED output.

Color Detected	Detection ID	RGB LED Output
Red	1	Red LED ON
Blue	2	Blue LED ON
Green	3	Green LED ON
Yellow	4	Yellow LED ON
Purple	5	Purple LED ON

Figure 3 shows the collected data

6.0 Data Analysis

The collected experimental data demonstrates that the Gravity HuskyLens AI Vision Camera was capable of accurately detecting and classifying the predefined colors under the given operating conditions. All trained colors, namely red, blue, green, yellow, and purple, were successfully recognized and assigned to their corresponding identification (ID) values ranging from 1 to 5. The HuskyLens consistently transmitted the correct ID information to the Arduino Uno through UART serial communication, allowing the microcontroller to process the visual data and generate the appropriate RGB LED response in real time.

The results also confirm that the communication interface between the HuskyLens AI camera and the Arduino Uno operated reliably without noticeable transmission errors or signal interruptions. Each detected color activated the corresponding RGB LED output accurately, indicating that the PWM-based LED control logic and serial data processing algorithm were implemented correctly. Furthermore, the response time between object detection and LED activation was observed to be fast and stable, demonstrating effective real-time system performance.

In addition, the experiment showed that the HuskyLens color recognition feature provided reliable detection performance after proper AI training and calibration. No major incorrect detections or false classifications were observed throughout the testing process, suggesting that the onboard machine learning algorithm was effective in distinguishing the trained colors. However, it was also observed that environmental factors such as ambient lighting intensity, shadows, and object positioning could potentially influence detection consistency if operating conditions were significantly altered.

Overall, the analysis confirms the successful integration of AI-assisted machine vision sensing, embedded microcontroller processing, UART communication, and real-time actuator feedback within a compact mechatronic system. The developed setup demonstrates strong potential for practical engineering applications including automated sorting systems, intelligent indicator devices, robotic inspection systems, and smart automation platforms requiring reliable color-based object recognition.

7.0 Results

The experiment was successfully conducted to evaluate the performance of the Gravity HuskyLens AI Vision Camera in detecting trained colors and generating corresponding RGB LED outputs through the Arduino Uno microcontroller platform. Throughout the testing process, the system consistently identified all five predefined colors, namely red, blue, green, yellow, and purple, with each color accurately associated with its assigned identification (ID) value.

When a trained color object was presented within the detection range of the HuskyLens camera, the system successfully recognized the object and transmitted the corresponding ID data to the Arduino Uno via UART serial communication. The Arduino then processed the received data and activated the RGB LED module using Pulse Width Modulation (PWM) control to display

the matching output color in real time. The LED indication accurately reflected the detected color during every testing cycle, confirming correct operation of both the AI vision sensing system and the embedded control logic.

The experimental results also demonstrated stable communication between the HuskyLens AI Vision Camera and the Arduino Uno without significant delays, communication failures, or incorrect detections during operation. The system maintained reliable performance under repeated testing conditions, indicating strong consistency and responsiveness in real-time color recognition applications. Overall, the experiment successfully achieved all intended objectives and demonstrated the effectiveness of integrating AI-based machine vision sensing with embedded microcontroller systems for intelligent color detection and visual output control.

8.0 Discussion

The experimental results demonstrate that the Gravity HuskyLens AI Vision Camera is capable of accurately detecting and classifying multiple predefined colors when properly trained under suitable operating conditions. Throughout the experiment, the system consistently recognized each trained color and assigned the correct identification (ID) value, which was then successfully processed by the Arduino Uno to generate the corresponding RGB LED output. This confirms that the integration between the AI vision sensor, embedded microcontroller platform, and actuator-based output system operated effectively and reliably in real time. The fast response of the RGB LED also indicates that the UART serial communication between the HuskyLens and Arduino Uno provided stable and efficient data transmission throughout the experiment.

One significant observation obtained from this experiment is the advantage of utilizing an AI-based machine vision sensor compared to conventional color sensing devices. Traditional color sensors typically rely on reflected light intensity measurements and often require calibration adjustments under varying environmental conditions. In contrast, the HuskyLens internally performs image acquisition, feature extraction, and color classification using embedded AI algorithms, significantly simplifying the implementation process while reducing the need for external image processing software or high-performance computing hardware. This allows the system to operate in a more flexible and user-friendly manner, especially in applications involving multiple object shapes, varying object orientations, and dynamic environments.

However, the experiment also revealed that environmental conditions remain an important factor influencing system performance. Variations in ambient lighting intensity, shadows, reflections, and background colors may affect the consistency and accuracy of color recognition if the operating environment changes significantly. As a result, proper AI training, stable illumination conditions, and suitable object positioning are essential to maintain reliable detection performance. In addition, careful calibration during the training process is critical because inaccurate or incomplete training data may lead to false detection, incorrect classification, or unstable system responses during operation.

Overall, this experiment successfully demonstrates the practical integration of AI-assisted machine vision sensing with embedded system control within a compact mechatronic platform.

The developed system highlights the potential application of intelligent vision-based sensing technologies in engineering fields such as automated product sorting, robotic inspection systems, smart traffic control systems, industrial quality monitoring, and interactive human-machine interface (HMI) applications. Furthermore, the experiment provides valuable practical exposure to embedded programming, UART communication, machine vision sensing, real-time signal processing, and actuator control integration within modern automation systems.

9.0 Conclusion

In conclusion, this experiment successfully demonstrated the implementation of an AI-based color detection and analysis system using the Gravity HuskyLens AI Vision Camera integrated with an Arduino Uno microcontroller platform. The developed system was capable of accurately detecting and classifying predefined colors in real time while generating the corresponding visual feedback through the RGB LED module. The experimental results confirmed that the communication between the HuskyLens AI camera and the Arduino Uno through UART serial communication operated reliably and efficiently throughout the testing process.

The experiment also demonstrated the effectiveness of AI-assisted machine vision sensing compared to conventional color detection approaches, as the HuskyLens was able to perform onboard image processing and color classification without requiring external software or complex computational systems. In addition, the integration of Pulse Width Modulation (PWM)-based RGB LED control further validated the successful coordination between sensor input, data processing, and actuator output within the embedded system.

Although the system achieved stable and accurate performance under controlled conditions, the experiment also highlighted the importance of proper AI training, calibration procedures, and stable lighting environments in maintaining reliable detection accuracy. Environmental factors such as illumination intensity, shadows, and object positioning can significantly influence system performance if not properly controlled.

Overall, this laboratory activity provided valuable practical exposure to AI vision sensing, embedded system programming, serial communication interfacing, real-time signal processing, and mechatronic system integration. The developed prototype demonstrates strong potential for practical engineering applications involving intelligent automation, robotic inspection, automated sorting systems, and smart interactive devices requiring efficient real-time color recognition and response capabilities.

10.0 Recommendations

For future improvements, several enhancements can be implemented to increase the reliability, accuracy, and practical usability of the AI-based color detection system. One important improvement involves optimizing the system performance under varying environmental lighting conditions. Although the HuskyLens AI Vision Camera demonstrated accurate color recognition in a controlled environment, changes in ambient brightness, shadows, reflections,

and background illumination may still influence detection consistency and classification accuracy.

To reduce these effects, a controlled lighting mechanism such as dedicated white LED illumination or an enclosed detection chamber can be incorporated to provide stable and uniform lighting conditions during operation. This would minimize external lighting interference and improve the repeatability of the color recognition process, especially in industrial or outdoor applications where environmental conditions are less predictable.

In addition, the system can be further enhanced by integrating more advanced AI training datasets involving additional colors, object patterns, or multiple object classifications. The implementation of filtering algorithms and adaptive thresholding techniques may also improve detection robustness under dynamic operating conditions. Furthermore, integrating additional output devices such as buzzers, LCD displays, relays, or robotic actuators could expand the functionality of the system for applications such as automated sorting systems, robotic inspection platforms, smart manufacturing processes, and intelligent human–machine interface (HMI) systems.

Lastly, future developments may explore wireless communication technologies such as Wi-Fi, Bluetooth, or IoT-based cloud monitoring to enable remote monitoring and data transmission capabilities. These enhancements would transform the current prototype into a more intelligent, scalable, and industry-oriented mechatronic automation system.

11.0 References

1. HuskyLens Official Repository. *HUSKYLENS Arduino Library*. Available at: <https://github.com/HuskyLens/HUSKYLENSArduino/>
2. DFRobot. *Gravity HuskyLens AI Vision Sensor Documentation*. Available at: https://wiki.dfrobot.com/HUSKYLENS_AI_Camera_SKU_SEN0305_SEN0336
3. Arduino Documentation. *Arduino Uno Rev3*. Available at: <https://docs.arduino.cc/hardware/uno-rev3/>

12.0 Acknowledgements

Special thanks to Dr. ZULKIFLI BIN ZAINAL ABIDIN and Dr. WAHJU SEDIONO for their guidance and support during this experiment.

13.0 Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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